claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-abd7acb156b1cba96.jsonl`
- SHA-256: `e4a4451030c73da5c4449a372df3773598f88066e77861526648e8a2446ad43f`
- Source modified: `2026-06-06T17:37:27+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:36:23.820Z)

Adversarial Claim Verifier (voter 2/3)

```
Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.
```

Research question

```
Research alternative methods to automatically detect rally start/end segments (temporal
segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST
pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach
or whether other methods are competitive or complementary. Keep it focused ("mini" research):
prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA
needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
```

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"HitNet with constrained-optimization post-processing reaches high hit-detection accuracy (94.3% recall, 97.2% precision, 0.946 F1), far above a derivative/velocity-threshold baseline on the shuttle trajectory (0.699 F1) ? direct evidence that a learned pose+trajectory hit model beats the velocity/threshold heuristics our current WASB-window approach relies on."

Source: https://ar5iv.labs.arxiv.org/html/2204.01899 (primary)
Supporting quote: "HitNet + optimization: 94.3% recall, 76.4% accuracy, 97.2% precision, 0.946 f1 ... Derivative-based: 65.7% recall, 53.8% accuracy, 74.7% precision, 0.699 f1"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:36:26.381Z)

I'll verify this claim by examining the primary source and checking the specific numbers and their interpretation.

3. assistant (2026-06-06T17:36:30.297Z)

I'll fetch the primary source to verify the exact numbers and the claim's interpretation.

4. user (2026-06-06T17:36:41.868Z)

MonoTrack Paper Analysis

(1) Performance Numbers (Table 1):

- **HitNet + optimization**: Recall 94.3%, Accuracy 89.7%, Precision 94.9%, F1 0.946
- **Derivative-based**: Recall 65.7%, Accuracy 53.8%, Precision 74.7%, F1 0.699

(2) Sport: Badminton

(3) HitNet Detection: HitNet detects "hit" events?the moments when a player strikes the shuttlecock, which are used to segment rallies into individual shots.

(4) HitNet Inputs: Both pose and trajectory. The paper states: "court layout, the poses, and the shuttlecock location play an important role in identifying hits." The architecture uses "pixel coordinates of the court corners, the two players' poses, and the location of the shuttle."

(5) Title & Dataset:
- **Title:** "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video"
- **Dataset:** Based on the public TrackNetV2 dataset containing 77k annotated frames from 26

singles matches

(6) Accuracy Caveat: The 76.4% accuracy for "HitNet (All)" without optimization appears low relative to its F1 of 0.866 because "accuracy" here uses a specialized metric: "only look at frames where hits occurred." This weighted evaluation explains the discrepancy?precision (97.2%) is very high, but recall (78.1%) is lower, producing an F1 score higher than the reported accuracy.

5. assistant (2026-06-06T17:36:45.000Z)

Interesting. The fetched numbers differ from the claim's quote. Let me look more carefully at the exact table to reconcile the discrepancy between the claim's quoted numbers and what's actually in the paper.

6. user (2026-06-06T17:36:52.241Z)

Hit Detection Results Table

Here is the exact table from the paper:

Table 1: "Comparison of HitNet over baselines and ablation study shows that HitNet benefits from all input features with the optimization-based postprocessing."

Method	Recall	Accuracy	Precision	F1
Derivative-based	65.7%	53.8%	74.7%	0.699
RF	65.1%	57.4%	83.0%	0.730
HitNet (Shuttle)	70.2%	68.8%	97.4%	0.815
HitNet (Shuttle+Pose)	73.9%	75.6%	97.1%	0.850
HitNet (All)	78.1%	76.4%	97.2%	0.866
HitNet+optimization	94.3%	89.7%	94.9%	0.946

Regarding accuracy definition: The paper states: "To measure the accuracy, recall, and precision of the models, we only look at frames where hits occurred. If we were to include all frames, then a trivial detector outputting no hit for all frames would get close to 100% accuracy."

The second row of your question (94.3% recall, 89.7% accuracy, 94.9% precision) is the correct one for HitNet+optimization.

7. assistant (2026-06-06T17:36:55.739Z)

I found a significant problem. The claim conflates numbers from two different rows. Let me verify this is the MonoTrack paper and check whether the claim's framing about "HitNet beats velocity/threshold heuristics our WASB-window approach relies on" is valid.

8. user (2026-06-06T17:37:02.161Z)

Web search results for query: "MonoTrack badminton HitNet shuttle trajectory hit detection arxiv 2204.01899"

Links: [{"title":"[2204.01899] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://ar5iv.labs.arxiv.org/html/2204.01899"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Request PDF","url":"https://www.researchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://openaccess.thecvf.com/content/CVPR2022W/CVSports/papers/Liu_MonoTrack_Shuttle_Trajectory_Reconstruction_From_Monocular_Badminton_Video_CVPRW_2022_paper.pdf"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"},{"title":"BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports","url":"https://arxiv.org/html/2502.21085v4"},{"title":"(PDF) MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://www.researchgate.net/publication/359757180_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"},{"title":"[PDF] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Semantic Scholar","url":"https://www.semanticscholar.org/paper/MonoTrack:-Shuttle-trajectory-reconstruction-from-Liu-Wang/8ac983e89347fb61373f1e6c659913a99a56a3d4"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | IEEE Conference Publication | IEEE Xplore","url":"https://ieeexplore.ieee.org/iel7/9856930/9856648/09857202.pdf"}]

Based on the search results, here's what I found about the paper with arxiv ID 2204.01899:

MonoTrack: Shuttle Trajectory Reconstruction from Monocular Badminton Video

The paper presents the first complete end-to-end system for the extraction and segmentation of 3D shuttle trajectories from monocular badminton videos, integrating badminton domain knowledge such as court dimension, shot placement, physical laws of motion, along with vision-based features such as player poses and shuttle tracking.

HitNet Hit Detection Component

The paper includes HitNet, a model that benefits from all input features with optimization-based postprocessing compared to baselines like derivative-based methods that detect hits by thresholding on trajectory derivatives and random forest-based classifiers.

The hit detection approach enforces the constraint that hits must alternate between opposing players, with optimization aiming to find a set of hits roughly maximizing the sum of confidence scores subject to constraints.

Key Contributions

The work improves state-of-the-art results on court recognition, 2D trajectory estimation, and hit recognition.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:37:08.168Z)

I have enough to render a verdict. Let me consolidate the findings.

The claim's quoted supporting numbers are: "HitNet + optimization: 94.3% recall, 76.4% accuracy, 97.2% precision, 0.946 f1".

The actual paper (MonoTrack, CVPRW 2022, arxiv 2204.01899) Table 1 shows two distinct rows:
- HitNet (All): 78.1% recall, 76.4% accuracy, 97.2% precision, 0.866 F1
- HitNet+optimization: 94.3% recall, 89.7% accuracy, 94.9% precision, 0.946 F1

The claim's quote is a **fabricated composite** ? it took recall/F1 from the "+optimization" row and accuracy/precision from the "HitNet (All)" row. No single row reads "94.3% recall, 76.4% accuracy, 97.2% precision, 0.946 f1."

Now I'll assess whether this misquote refutes the claim. I'll be precise about what is and isn't refuted.

10. user (2026-06-06T17:37:22.145Z)

Structured output provided successfully

11. assistant (2026-06-06T17:37:27.701Z)

I've completed the verification and submitted my structured output. The claim is **refuted**.

Key finding: the claim's supporting quote ("94.3% recall, 76.4% accuracy, 97.2% precision, 0.946 f1") is a fabricated composite that stitches together numbers from two different rows of MonoTrack's Table 1 ? recall/F1 come from the "HitNet+optimization" row while accuracy/precision come from the separate "HitNet (All)" row. No single row in the source contains all four of those numbers. Additionally, the interpretation overreaches by equating per-shot hit-event detection with the rally-segmentation task and ignoring that HitNet requires pose + court-homography inputs our current WASB-only pipeline lacks.

The directional core (learned pose+trajectory hit model beats a derivative baseline, 0.946 vs 0.699 F1) is real, but the claim as stated fails the misquote and overreach checks.